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A debugger is a tool used to test and debug other programs or target programs. This project is based on Atmega16A microcontroller (MCU), and is used to check the correct data received through universal synchronous/asynchronous receiver/transmitter (USART) interface on an LCD. It is a useful tool for debugging USART data for MCUs and related circuitry.

Generally, to check USART data of MCU, Rx and Tx pins are connected to COM port of the PC through voltage-level converters like MAX232. A program (for example, HyperTerminal) is then used to check correct data

transmission on the computer terminal or screen. But, it is still difficult to check/identify non-printable characters on the computer screen. These days, a COM port is not available on PCs and laptops, and more hardware setup is needed for USB to COM port conversion.

This simple and handy serial data debugger can solve the problem. In this project, instead of COM port two similar circuits are wired to transfer the serial data for testing.

Circuit and working

The circuit diagram of the serial data debugger is shown in Fig. 1. It is built

around ATmega16A MCU (IC1), 16x4 LCD (LCD1) and a few other components.

ATmega16A employs a 11.0592MHz crystal to have accurate baud rate. LCD1 is used as output/console.

USART frame size/format can be varied as per requirement by pressing such buttons as baud rate (S11), data bits (S10), parity (S9) and stop bits (S8). This frame detail will be displayed on the fourth line of LCD1. Use null switch S3 to enable or disable null values to be added to the buffer while receiving the data stream.

Two circuits (say, MCU1 and MCU2) of Fig. 1 are wired on a separate PCB. Their Tx and Rx pins are connected to each other. Serial data is sent by MCU2 through its Tx pin and received by Rx pin of MCU1. Serial data is written to a buffer in MCU1 for debugging received data. Maximum buffer size is 255 characters.

Each buffer position and its byte data occupy two columns of LCD1. So, LCD1 can display eight-character data of received buffer at a time.

First row of LCD1 displays position of the buffer in hexadecimal format, second row displays actual character, third row displays data in hexadecimal format and fourth row displays frame size of data being received.

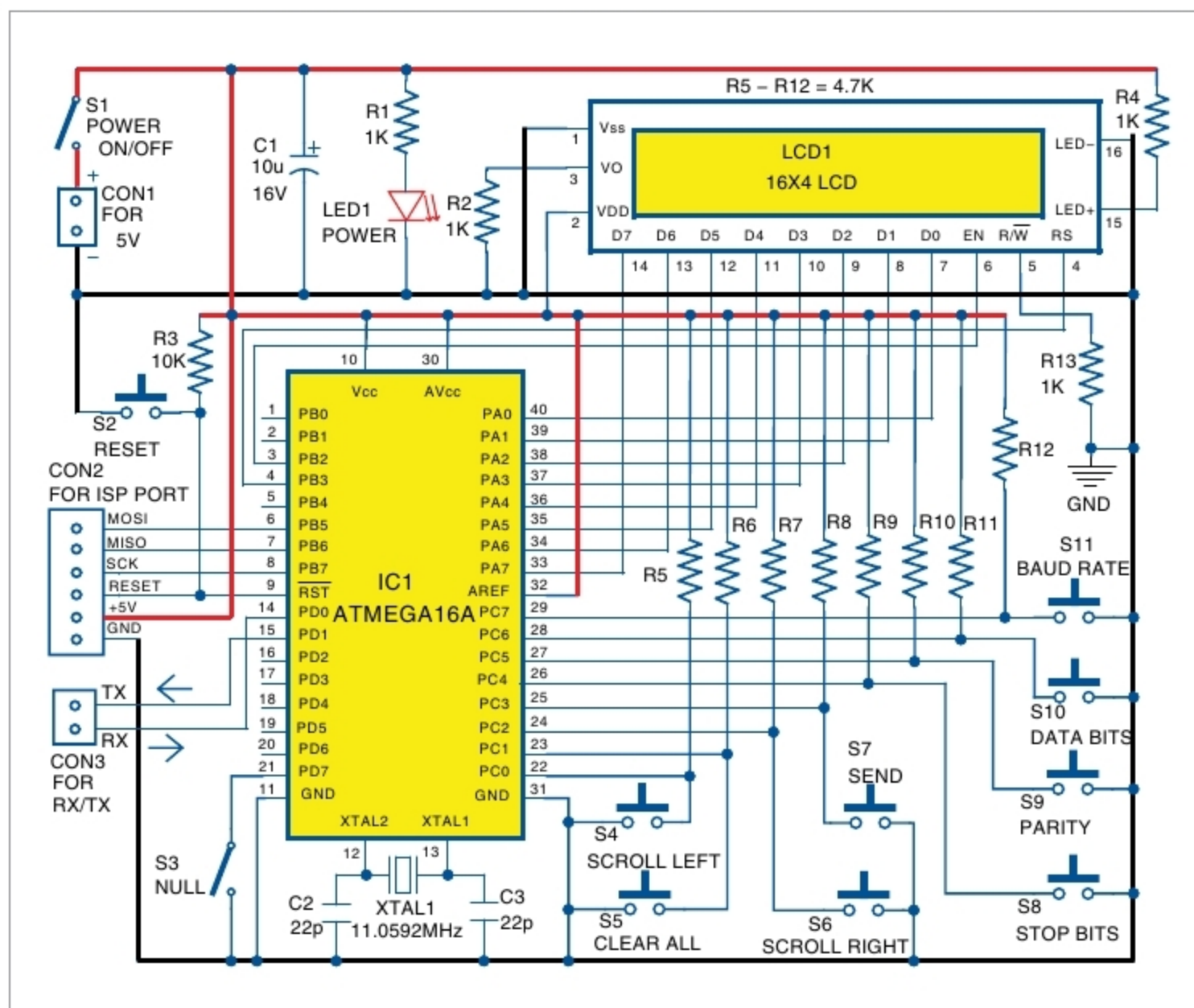


Fig. 1: Circuit diagram of the serial data debugger